

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER– III (New) EXAMINATION – WINTER 2019****Subject Code: 3130005****Date: 26/11/2019****Subject Name: Complex Variables and Partial Differential Equations****Time: 02:30 PM TO 05:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

		Marks
Q.1	(a) Find the real and imaginary parts of $f(z) = \frac{3i}{2+3i}$.	03
	(b) State De-Moivre's formula and hence evaluate $(1+i\sqrt{3})^{100} + (1-i\sqrt{3})^{100}$.	04
	(c) Define harmonic function. Show that $u(x, y) = \sinh x \sin y$ is harmonic function, find its harmonic conjugate $v(x, y)$.	07
Q.2	(a) Determine the Mobius transformation which maps $z_1 = 0, z_2 = 1, z_3 = \infty$ into $w_1 = -1, w_2 = -i, w_3 = 1$.	03
	(b) Define $\log z$, prove that $i^i = e^{-(4n+1)\frac{\pi}{2}}$.	04
	(c) Expand $f(z) = \frac{1}{(z-1)(z+2)}$ valid for the region (i) $ z < 1$ (ii) $1 < z < 2$ (iii) $ z > 2$.	07
	OR	
	(c) Find the image of the infinite strips (i) $\frac{1}{4} \leq y \leq \frac{1}{2}$ (ii) $0 < y < \frac{1}{2}$ under the transformation $w = \frac{1}{z}$. Show the region graphically.	07
Q.3	(a) Evaluate $\int_c (x - y + ix^2) dz$ where c is a straight line from $z = 0$ to $z = 1 + i$.	03
	(b) Check whether the following functions are analytic or not at any point, (i) $f(z) = 3x + y + i(3y - x)$ (ii) $f(z) = z^{3/2}$.	04
	(c) Using residue theorem, evaluate $\int_0^\infty \frac{dx}{(x^2+1)^2}$.	07
	OR	
Q.3	(a) Expand Laurent series of $f(z) = \frac{1-e^z}{z}$ at $z = 0$ and identify the singularity.	03
	(b) If $f(z) = u + iv$, is an analytic function, prove that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) Ref(z) ^2 = 2 f'(z) ^2$.	04
	(c) Evaluate the following: i. $\int_c \frac{z+3}{z-1} dz$ where c is the circle (a) $ z = 2$ (b) $ z = \frac{1}{2}$. ii. $\int_c \frac{\sin z}{\left(z-\frac{\pi}{4}\right)^3} dz$ where c is the circle $ z = 1$.	07

- Q.4** (a) Evaluate $\int_0^{2+4i} \operatorname{Re}(z) dz$ along the curve $z(t) = t + it^2$. **03**
- (b) Solve $x^2 p + y^2 q = (x + y)z$. **04**
- (c) Solve the equation $\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$ for the condition of heat along rod without radiation subject to the conditions (i) $\frac{\partial u}{\partial t} = 0$ for $x = 0$ and $x = l$; **07**
(ii) $u = lx - x^2$ at $t = 0$ for all x .
- OR**
- Q.4** (a) Solve $\frac{\partial^2 z}{\partial x^2} + 2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = e^{2x+3y}$. **03**
- (b) Solve $px + qy = pq$ using Charpit's method. **04**
- (c) Find the general solution of partial differential equation $u_{xx} = 9u_y$ using method of separation of variables. **07**
- Q.5** (a) Using method of separation of variables, solve $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$. **03**
- (b) Solve $z(xp - yq) = y^2 - x^2$. **04**
- (c) A string of length $L = \pi$ has its ends fixed at $x = 0$ and $x = \pi$. At time $t = 0$, the string is given a shape defined by $f(x) = 50x(\pi - x)$, then it is released. Find the deflection of the string at any time t . **07**
- OR**
- Q.5** (a) Solve $p^3 + q^3 = x + y$. **03**
- (b) Find the temperature in the thin metal rod of length l with both the ends insulated and initial temperature is $\sin^{\pi x}/l$. **04**
- (c) Derive the one dimensional wave equation that governs small vibration of an elastic string . Also state physical assumptions that you make for the system. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER– III EXAMINATION – SUMMER 2020****Subject Code: 3130005****Date: 27/10/2020****Subject Name: Complex Variables and Partial Differential Equations****Time: 02:30 PM TO 05:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

		Marks
Q.1	(a) If $u = x^3 - 3xy$ is find the corresponding analytic function $f(z) = u + iv$.	03
	(b) Find the roots of the equation $z^2 - (5 + i)z + 8 + i = 0$.	04
	(c) (i) Determine and sketch the image of $ z = 1$ under the transformation $w = z + i$.	03
	(ii) Find the real and imaginary parts of $f(z) = z^2 + 3z$.	04
Q.2	(a) Evaluate $\int_C (x^2 - iy^2) dz$ along the parabola $y = 2x^2$ from (1,2) to (2,8).	03
	(b) Find the bilinear transformation that maps the points $z = \infty, i, 0$ into $w = 0, i, \infty$.	04
	(c) (i) Evaluate $\oint_C \frac{e^{-z} dz}{z+1}$, where C is the circle $ z = 1/2$.	03
	(ii) Find the radius of convergence of $\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^{n^2} z^n$.	04
	OR	
	(c) (i) Find the fourth roots of -1 .	03
	(ii) Find the roots of $\log z = i \frac{\pi}{2}$.	04
Q.3	(a) Find $\oint_C \frac{1}{z^2} dz$, where $C : z = 1$.	03
	(b) For $f(z) = \frac{1}{(z-1)^2(z-3)}$, find Residue of $f(z)$ at $z=1$.	04
	(c) Expand $f(z) = \frac{1}{(z+2)(z+4)}$ in a Laurent series for the regions (i) $ z < 2$, (ii) $2 < z < 4$, (iii) $ z > 4$.	07
	OR	
Q.3	(a) Find $\oint_C \frac{z+4}{z^2+2z+5} dz$, where $C : z+1 = 1$.	03
	(b) Evaluate using Cauchy residue theorem $\int_C \frac{e^{2z}}{(z+1)^3} dz$; $C: 4x^2 + 9y^2 = 16$.	04
	(c) Expand $f(z) = \frac{1}{z(z-1)(z-2)}$ in Laurent's series for the regions (i) $ z < 1$, (ii) $1 < z < 2$, (iii) $ z > 2$.	07

- Q.4** (a) Solve $xp + yq = x - y$. **03**
- (b) Derive partial differential equation by eliminating the arbitrary constants a and b from $z = ax + by + ab$. **04**
- (c) (i) Solve the p.d.e. $2r + 5s + 2t = 0$. **03**
- (ii) Find the complete integral of $p^2 = qz$. **04**
- OR**
- Q.4** (a) Find the solution of $x^2p + y^2q = z^2$. **03**
- (b) Form the partial differential equation by eliminating the arbitrary function ϕ from $z = \phi\left(\frac{y}{x}\right)$. **04**
- (c) (i) Solve the p.d.e. $(D^2 - D'^2 + D - D')z = 0$. **03**
- (ii) Solve by Charpit's method $yzp^2 - q = 0$. **04**
- Q.5** (a) Solve $(2D^2 - 5DD' + 2D'^2)z = 24(y - x)$. **03**
- (b) Solve the p.d.e. $u_x + u_y = 2(x + y)u$ using the method of separation of variables. **04**
- (c) Find the solution of the wave equation $u_{tt} = c^2 u_{xx}$, $0 \leq x \leq \pi$ with the initial and boundary conditions $u(0, t) = u(\pi, t) = 0; t > 0$, $u(x, 0) = k(\sin x - \sin 2x), u_t(x, 0) = 0; 0 \leq x \leq \pi$. ($c^2 = 1$) **07**
- OR**
- Q.5** (a) Solve the p.d.e. $r + s + q - z = 0$. **03**
- (b) Solve $2u_x = u_t + u$ given $u(x, 0) = 4e^{-3x}$ using the method of separation of variables. **04**
- (c) Find the solution of $u_t = c^2 u_{xx}$ together with the initial and boundary conditions $u(0, t) = u(2, t) = 0; t \geq 0$ and $u(x, 0) = 10; 0 \leq x \leq 2$. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-III (NEW) EXAMINATION – WINTER 2020****Subject Code:3130005****Date:09/03/2021****Subject Name:Complex Variables and Partial Differential Equations****Time:10:30 AM TO 12:30 PM****Total Marks:56****Instructions:**

1. Attempt any FOUR questions out of EIGHT questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

		Marks
Q.1	(a) Show that the function $u = x^2 - y^2 + x$ is harmonic.	03
	(b) Find the fourth roots of -1 .	04
	(c) (i) Find and sketch the image of the region $ z > 1$ under the transformation $w = 4z$.	03
	(ii) Find the real and imaginary parts of $f(z) = z^2 + 3z$.	04
Q.2	(a) Evaluate $\int_0^{2+i} z^2 dz$ along the line $y = x/2$.	03
	(b) Determine the Mobius transformation that maps $z = 0, 1, \infty$ onto $w = -1, -i, 1$ respectively.	04
	(c) (i) Evaluate $\oint_C \frac{e^{-z}}{z+1} dz$, where C is the circle $ z = 1/2$.	03
	(ii) For which values of z does the series $\sum_{n=0}^{\infty} \frac{z^n}{n!}$ convergent?	04
Q.3	(a) Evaluate $\oint_C \frac{dz}{z^2}$ where C is a unit circle.	03
	(b) Find the residue $\text{Res}(f(z), 2i)$ of the function $f(z) = \frac{z^2 - 2z}{(z+1)^2(z^2 + 4)}$.	04
	(c) Expand $f(z) = \frac{1}{(z+1)(z-2)}$ in Laurent's series in the region (i) $ z < 1$, (ii) $1 < z < 2$, (iii) $ z > 2$.	07
Q.4	(a) Evaluate $\oint_C \frac{5z+7}{z^2+2z-3} dz$, where $C: z-2 = 2$.	03
	(b) Evaluate by using Cauchy's Residue Theorem $\int_C \frac{5z-2}{z(z-1)} dz$, $C: z = 2$.	04
	(c) Find Laurent's series that represent $f(z) = \frac{1}{z(z-1)}$ in the region (i) $0 < z < 1$, (ii) $0 < z-1 < 1$.	07

- Q.5** (a) Solve $xp + yq = x - y$. **03**
 (b) Derive p.d.e. from $z = ax + by + ab$ by eliminating a and b . **04**
 (c) (i) Solve $(D^3 - 4D^2D' + 4DD'^2)z = 0$. **03**
 (ii) Solve $pq = k$, where k is a constant. **04**
- Q.6** (a) Solve $zp + yq = x$. **03**
 (b) Form the p.d.e. by eliminating ϕ from $x + y + z = \phi(x^2 + y^2 + z^2)$. **04**
 (c) (i) Solve $(D^2 + DD' + D' - 1)z = \sin(x + 2y)$. **03**
 (ii) Solve $zpq = p + q$ by Charpit's method. **04**
- Q.7** (a) Solve $\frac{\partial^3 z}{\partial x^3} - 2\frac{\partial^3 z}{\partial x^2 \partial y} = 2e^{2x}$. **03**
 (b) Solve the p.d.e. $u_x = 4u_y$, $u(0, y) = 8e^{-3y}$. **04**
 (c) A string of length l is initially at rest in its equilibrium position and motion is started by giving each of its points a velocity kx for $0 \leq x \leq l/2$ and $k(l-x)$ for $l/2 \leq x \leq l$. Find the displacement $u(x, t)$. **07**
- Q.8** (a) Solve $DD''(D - 2D' - 3)z = 0$. **03**
 (b) Solve the pde $u_{xx} = 16u_{xy}$. **04**
 (c) A bar of length $2m$ is fully insulated along its sides. It is initially at a uniform temperature of $10^\circ C$ and at $t = 0$ the ends are plunged into ice and maintained at a temperature of $0^\circ C$. Determine an expression for the temperature at a point P at a distance x from one end at any subsequent time t seconds after $t = 0$. **07**

Seat No.: _____

Enrolment No. _____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER-III (NEW) EXAMINATION – SUMMER 2021

Subject Code: 3130005

Date: 05/10/2021

Subject Name: Complex Variables and Partial Differential Equations

Time: 10:30 AM TO 01:00 PM

Total Marks: 70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

MARKS

- Q.1**
- | | | |
|-----|---|-----------|
| (a) | Find the fourth roots of -1 . | 03 |
| (b) | Find the real and imaginary parts of $(-1-i)^7 + (-1+i)^7$. | 04 |
| (c) | (i) Find the analytic function whose real part is $u = x^2 - y^2$. | 04 |
| | (ii) Show that $u = \frac{x}{x^2 + y^2}$ is harmonic. | 03 |

- Q.2**
- | | | |
|-----|--|-----------|
| (a) | Find the image in the w – plane of the circle $ z - 3 = 2$ in the z – plane under the inversion mapping $w = \frac{1}{z}$. | 03 |
| (b) | Evaluate $\int_C (x^2 - iy^2) dz$ along the parabola $y = 2x^2$ from $(1, 2)$ to $(2, 8)$. | 04 |
| (c) | State Cauchy's Integral Formula. Evaluate $\oint_C \frac{e^{2z}}{(z+1)^4} dz$, where C is the circle $ z = 2$. | 07 |

OR

- | | | |
|-----|--|-----------|
| (c) | (i) Evaluate $\oint_C \frac{e^{-z}}{z+1} dz$, where C is the circle $ z = \frac{1}{2}$. | 04 |
| | (ii) Determine and sketch the image of $ z = 1$ under the transformation $w = z + i$. | 03 |

- Q.3**
- | | | |
|-----|---|-----------|
| (a) | Find the radius of convergence of $\sum_{n=1}^{\infty} (3+4i)^n z^n$. | 03 |
| (b) | Find the residues of the function $f(z) = \frac{1}{(z-1)^2(z-3)}$ at each of its poles in the finite z – plane. | 04 |
| (c) | Expand $f(z) = \frac{1}{(z+1)(z+3)}$ in a Laurent series for the regions (i) $ z < 1$, (ii) $1 < z < 3$, (iii) $ z > 3$. | 07 |

OR

- Q.3**
- | | | |
|-----|--|-----------|
| (a) | Find the bilinear transformation which transforms $z = 2, 1, 0$ into $w = 1, 0, i$. | 03 |
| (b) | Evaluate $\int_C \frac{e^{2z}}{(z+1)^3} dz$ by using Cauchy Residue Theorem, where C is the ellipse $4x^2 + 9y^2 = 16$. | 04 |

- (c) Find the Laurent series for the function $f(z) = \frac{1}{z(1-z)}$ in the region (i) $|z+1| < 1$ 07
, (ii) $1 < |z+1| < 2$, (iii) $|z+1| > 2$.
- Q.4** (a) Find the real and imaginary parts of $f(z) = z^3 + 3z$. 03
(b) Derive partial differential equation by eliminating arbitrary constants a and b from $z = (x^2 + a)(y^2 + b)$. 04
(c) (i) Solve $x^2 p + y^2 q = z^2$. 03
(ii) Solve $q = 3p^2$ by Charpit's method. 04
- OR**
- Q.4** (a) Show that the function $f(z) = xy + iy$ is continuous everywhere but not analytic. 03
(b) Solve $yq - xp = z$. 04
(c) (i) Find the complete integral of $q = pq + p^2$. 03
(ii) Solve $px + qy = pq$ by Charpit's method. 04
- Q.5** (a) Solve $r - 2s + t = \sin(2x + 3y)$. 03
(b) Solve $(D^2 - D'^2 + D - D')z = 0$. 04
(c) Solve $u_x = 4y_y$, $u(0, y) = 8e^{-3y}$ by the method of separation of variables. 07
- OR**
- Q.5** (a) Solve $(DD' + D - D' - 1)z = xy$. 03
(b) Solve $(D^2 + 3DD' + 2D'^2)z = x + y$. 04
(c) A tightly stretched string with fixed end points $x = 0$, $x = l$ is initially at rest in its equilibrium position. If it is set vibrating giving each point a velocity $\lambda x(l - x)$, find the displacement $u(x, t)$. 07

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-III (NEW) EXAMINATION – WINTER 2021****Subject Code:3130005****Date:17-02-2022****Subject Name:Complex Variables and Partial Differential Equations****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Represent $z = 7i$ into polar form and find the argument of z and the principal value of the argument of z .	03
	(b) State De Moivre's theorem. Find and plot all roots of $(1+i)^{\frac{1}{3}}$ in the complex plane.	04
	(c) Using the method of separation of variables, solve $4\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 3u$ and $u = e^{-5y}$ when $x = 0$.	07
Q.2	(a) Define an analytic function. Write the necessary and sufficient condition for function $f(z)$ to be analytic. Show that $f(z) = z ^2$ is nowhere analytic.	03
	(b) Define the Mobious transformation. Determine the bilinear transformation which mapping the points $0, \infty, i$ onto $\infty, 2, 0$.	04
	(c) Attempt the following.	
	(i) Define the harmonic function. Show that $u = x^2 - y^2 + x$ is harmonic and find harmonic conjugate of u .	04
	(ii) Show that $f(z) = \begin{cases} \frac{\text{Im}(z)}{ z } & ; z \neq 0 \\ 0 & ; z = 0 \end{cases}$ is not continuous at $z = 0$	03
	OR	
	(c) Attempt the following.	
	(i) Prove that $\cos^{-1} z = -i \ln(z + i\sqrt{1-z^2})$.	4
	(ii) Find the values of $\text{Re } f(z)$ and $\text{Im } f(z)$ at the point $7+2i$, where $f(z) = \frac{1}{1-z}$.	3
Q.3	(a) Evaluate $\int_C \bar{z} dz$, where C is the right- half of the circle $ z = 2$ and hence show that $\int_C \frac{dz}{z} = \pi i$.	03
	(b) Expand $f(z) = \sin z$ in a Taylor series about $z = \frac{\pi}{4}$ and write the Maclaurin series for e^{-z} .	04
	(c) Write the Cauchy integral theorem and Cauchy integral formula and hence evaluate:	07
	(i) $\oint_C \frac{e^z}{(z-1)(z-3)} dz; C: z = 2$. (ii) $\oint_C e^{\bar{z}} dz; C: z = 3$.	

OR

- Q.3** (a) Evaluate $\oint_C \frac{e^z}{z+i} dz$, where $C: |z-1|=1$. **03**
- (b) Develop the following functions into Maclaurin series: (i) $\cos^2 z$ (ii) $e^z \cos z$. **04**
- (c) Evaluate $\int_C \operatorname{Re}(z^2) dz$, where C is the boundary of the square with vertices $0, i, 1+i, 1$ in the clockwise direction. **07**
- Q.4** (a) Define the singular points of $f(z)$. Find the singularity of $f(z)$ and classify as pole, essential singularity or removable singularity. where $f(z) = \frac{1-e^z}{z}$. **03**
- (b) State the Cauchy residue theorem. Find the residue at its poles of $f(z) = \frac{z^2}{(z-1)^2(z+2)}$ and hence evaluate $\oint_C f(z) dz$, $C: |z|=3$. **04**
- (c) Determine the Laurent series expansion of $f(z) = \frac{1}{(z+2)(z+4)}$ valid for regions
(i) $|z| < 2$
(ii) $2 < |z| < 4$
(iii) $|z| > 4$ **07**

OR

- Q.4** (a) Solve $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial y^2} = e^{x+2y}$. **03**
- (b) Form a partial differential equation by eliminating the arbitrary functions from the equations $z = f(x+ay) + \phi(x-ay)$. **04**
- (c) Show that $\int_{-\infty}^{\infty} \frac{dx}{(x^2+1)(x^2+4)} = \frac{\pi}{6}$. **07**
- Q.5** (a) Form a partial differential equation by eliminating the arbitrary function from $u = f\left(\frac{x}{y}\right)$. **03**
- (b) State the Lagrange's linear partial differential equation of first order and hence solve $x(y-z)p + y(z-x) = z(x-y)$ **04**
- (c) Using the method of separation of variables, solve $\frac{\partial u}{\partial x} = 4 \frac{\partial u}{\partial y}$ where $u(0, y) = 8e^{-3y}$. **07**

OR

- Q.5** (a) Obtain the general solution of $p + q^2 = 1$. **03**
- (b) Solve by Charpit's method: $px + qy = pq$. **04**
- (c) Find the solution of the wave equation $u_{tt} = c^2 u_{xx}$,
 $0 \leq x \leq L$ satisfying the condition $u(0, t) = u(L, t) = 0$, $u_t(x, 0) = 0$,
 $u(x, 0) = \frac{\pi x}{L}$, $0 \leq x \leq L$. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER– III (NEW) EXAMINATION – SUMMER 2022****Subject Code:3130005****Date:11-07-2022****Subject Name:Complex Variables and Partial Differential Equations****Time:02:30 PM TO 05:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	Marks
Q.1 (a) Express the complex number $-\sqrt{3} - i$ in polar form.	03
(b) Use De Moivre's theorem and find $\sqrt[3]{64i}$.	04
(c) Verify that $u = 2x - x^3 + 3xy^2$ is harmonic in the whole complex plane and find its harmonic conjugate function $v(x, y)$.	07
Q.2 (a) Discuss Continuity of the function $f(z)$ at the origin:	03
$f(z) = \begin{cases} \frac{\text{Im}(z)}{z}, & z \neq 0 \\ 0 & z = 0 \end{cases}$	
(b) 1) Define $\text{Log}(x + iy)$ 2) Determine $\text{Log}(-1 + i)$ 3) Determine all values of $\log(1 + i)$	04
(c) Find the image of the circle $ z + i = 2$ under the transformation $w = \frac{1}{z}$. Also, show the regions graphically.	07
OR	
(c) Check whether the function $f(z) = \sin z$ is analytic or not. If so, find its derivative.	07
Q.3 (a) Evaluate $\oint_C \frac{\sin z}{(z-\pi)^2} dz$, where C is the circle $ z = 4$	03
(b) Find the Laurent's series that represent $f(z) = \frac{1}{(z-2)(z-3)}$ in the region $2 < z < 3$.	04
(c) Find the residues of the function $f(z) = \frac{z}{(z+1)^2(z^2-4)}$ at its poles.	07
OR	
Q.3 (a) Evaluate $\int_0^{2+i} z^2 dz$ along the line $y = x$	03
(b) Evaluate $\oint_C \frac{3z+4}{z^2+2z-3} dz$, where C is $ z = 2$	04
(c) Using Residue theorem, evaluate the following Integral:	07
$\int_0^{2\pi} \frac{d\theta}{5 - 3 \sin \theta}$	
Q.4 (a) Expand $f(z) = \frac{\sin z}{z^4}$ in Laurent's series about $z = 0$ and identify the singularity.	03
(b) Solve: $\frac{\partial^2 z}{\partial x^2} + 10 \frac{\partial^2 z}{\partial x \partial y} + 25 \frac{\partial^2 z}{\partial y^2} = e^{3x+2y}$	04
(c) Solve $x^2 p + y^2 q = (x + y)z$	07

OR

- Q.4** (a) Find the fixed points of the transformation, $w = \frac{z-1}{z+1}$ **03**
(b) Solve: $\frac{\partial^2 z}{\partial x^2} + 2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = \cos x$ **04**
(c) Solve: $\frac{\partial^2 z}{\partial x \partial y} = \sin x \sin y$, given that $\frac{\partial z}{\partial y} = -2 \sin y$ when $x=0$ and $z=0$ **07**
when y is an odd multiple of $\frac{\pi}{2}$

- Q.5** (a) Solve $xp + yq = 3z$ **03**
(b) Solve by the method of separation of variables $\frac{\partial u}{\partial x} = 4 \frac{\partial u}{\partial y}$, where **04**
 $u(0, y) = 8e^{-3y}$
(c) A tightly stretched string with fixed end points at $x = 0$ and $x = 10$ is **07**
initially given by the deflection $f(x) = kx(10 - x)$. If it is released from
this position, then find the deflection of the string.

OR

- Q.5** (a) Find complete and singular solution of $z = px + qy + pq$ **03**
(b) Using Charpit's method, solve $q = 3p^2$. **04**
(c) A rod of 30 cm long has its ends A and B are kept at 20°C and 80°C **07**
respectively until steady state conditions prevail. The temperature at each
end is suddenly reduced to 0°C and kept so. Find the resulting temperature
 $u(x, t)$ from the end A .

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER- III(NEW) EXAMINATION – WINTER 2022****Subject Code: 3130005****Date: 20-02-2023****Subject Name: Complex Variables and Partial Differential Equations****Time: 02:30 PM TO 05:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) If $Z = \frac{2+6\sqrt{3}i}{5+\sqrt{3}i}$ then find its modulus value and the principal value of its argument.	03
	(b) Find all the values of $(1 - i)^{2/3}$	04
	(c) Solve $\frac{\partial^2 z}{\partial x^2} - 3 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} = \cos(x + 2y)$	07
Q.2	(a) Give examples of functions $f(z)$ and $g(z)$ which are analytic everywhere in the complex plane $\mathbb{C}$ such that $f(z)$ is never zero and $g(z) = 0$ if and only if $z = i$.	03
	(b) Discuss the continuity of $f(z) = \frac{\operatorname{Re}(z^2)}{ z ^2}, \quad z \neq 0,$ at $z = 0$ if $f(0) = 0$.	04
	(c) Define Mobius transformation. Find the mobius transformation which maps the points $z = 0, -i, -1$ to $w = i, 1, 0$ respectively.	07
	OR	
	(c) Define Harmonic function and show that $u(x, y) = 2x - x^3 + 3xy^2$ is harmonic. State what are Cauchy Riemann equations and use it to find the harmonic conjugate of the given function $u(x, y)$.	07
Q.3	(a) State: (i) Liouville Theorem and (ii) Cauchy-Goursat Theorem.	03
	(b) Find the radius of convergence of the power series $\sum_{n=1}^{\infty} \frac{n!}{n^n} z^n$ using the appropriate formula.	04
	(c) Evaluate $\int_C \bar{z} dz$ where C is along the sides of the triangle having vertices $z = 0, 1, i$.	07
	OR	
Q.3	(a) Expand $\ln(1 + z)$ as a Taylor series about $z = 0$ for $ z < 1$.	03
	(b) Without computing the integral, show that $\left \int_C \frac{e^z}{z+1} dz \right \leq \frac{8\pi e^4}{3}$ where C denotes the circle centered at origin and with radius 4.	04
	(c) State the Generalized Cauchy integral formula and use it to compute $\int_C \frac{z + e^z}{(z-1)^3} dz$ where C is $ z = 2$.	07

- Q.4 (a)** Identify the type of singularities of the function $f(z) = (z^2 - z^6)^{-1}$. **03**
- (b)** Using the Cauchy residue theorem, compute $\int_C \frac{1}{(z-1)^2(z-3)} dz$ where C is $|z| = 4$. **04**
- (c)** Solve $(x^2 + 2y^2)p - xyq = xz$. **07**
- OR**
- Q.4 (a)** Form the partial differentiation equation by eliminating the arbitrary constants from $(x - a)(x - b) - z^2 = x^2 + y^2$. **03**
- (b)** Find the complete integral (complete solution) of $q^2 = z^2 p^2 (1 - p^2)$. **04**
- (c)** Find the Laurent's series of $f(z) = \frac{3}{(z-2)(z+1)}$ in the regions: **07**
- (i) $|z| < 1$ and (ii) $1 < |z| < 2$.
- Q.5 (a)** Solve $25 \frac{\partial^2 z}{\partial x^2} - 40 \frac{\partial^2 z}{\partial x \partial y} + 16 \frac{\partial^2 z}{\partial y^2} = 0$. **03**
- (b)** Find the complete integral (complete solution) of $p^2 - q^2 = x - y$. **04**
- (c)** Solve $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 2(x + y)u$ using the method of separation of variables. **07**
- OR**
- Q.5 (a)** Solve $\frac{\partial^2 z}{\partial x^2} + 10 \frac{\partial^2 z}{\partial x \partial y} + 25 \frac{\partial^2 z}{\partial y^2} = e^x e^{-y}$. **03**
- (b)** Using the contour integration, show that the value of the improper integral $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$ is equal to π . **04**
- (c)** A tightly stretched string with fixed end points $x = 0$ and $x = 1$ in the shape defined by $y = x(1 - x)$ is released from this position of rest. Find $y(x, t)$ using the wave equation $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-III(NEW) EXAMINATION – SUMMER 2023****Subject Code:3130005****Date:24-07-2023****Subject Name: Complex Variables and Partial Differential Equations****Time: 02:30 PM TO 05:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

MARKS

- Q.1** (a) Determine an analytic function whose real part is $e^{2x}(x\cos 2y - y\sin 2y)$ **03**
 (b) Solve the equation $z^2 + (2i - 3)z + 5 - i = 0$ **04**
 (c) Show that $u(x, y) = 2x - x^3 + 3xy^2$ is harmonic and **07**
 Find a harmonic conjugate of $u(x, y)$

- Q.2** (a) Evaluate $\int_C |z|^2 dz$ around the square with vertices at (0,0), (1,0), (1,1), (0,1). **03**
 (b) Expand $f(z) = \frac{1}{(z+2)(z+4)}$ valid for the following regions **04**
 (i) $|z| < 2$ (ii) $2 < |z| < 4$
 (c) (i) Evaluate $\int_C \frac{zdz}{(z-1)(z-2)}$ where C is the circle $|z| = \frac{1}{2}$ **03**
 (ii) Evaluate $\int_C \frac{dz}{z^2 - 7z + 12}$ where C is the circle $|z| = 3.5$ **04**

OR

- (c) Define mobius transformation. Determine the mobius transformation which maps $z_1 = 0, z_2 = 1, z_3 = \infty$ onto $w_1 = -1, w_2 = -i, w_3 = 1$ **07**
Q.3 (a) Find and plot the image of triangular region in the z-plane with vertices (0,0), (1,0), (0,1) under the transformation $w = (1 - i)z + 3$ **03**
 (b) Find the values of a and b such that the function $f(z) = x^2 + ay^2 - 2xy + i(bx^2 - y^2 + 2xy)$ is analytic. **04**
 (c) Determine the poles of the function $f(z) = \frac{z^2}{(z-1)^2(z+2)}$ and Residue at each pole. Hence evaluate $\int_C f(z)dz$ where C is the circle $|z| = 3$ **07**

OR

- Q.3** (a) Expand $f(z) = \frac{1-e^z}{z}$ in Laurent's series about $z = 0$. **03**
 (b) Find modulus and argument of **04**
 (i) $\frac{1+2i}{1-(1-i)^2}$ (ii) $\frac{(1+i)^2}{1-i}$
 (c) Evaluate (i) $\int_C \frac{3z^2+7z+1}{z+1} dz$ Where C is $|z| = \frac{1}{2}$ **03**
 (ii) $\int_C \frac{z^2+1}{z^2-1} dz$ Where C is $|z-1| = 1$ **04**

- Q.4** (a) Solve $yzp - xzq = xy$ **03**
 (b) Form partial differential equation by eliminating the arbitrary constants a and b from $z = axe^y + \frac{1}{2}a^2e^{2y} + b$ **04**
 (c) (i) Solve $25r - 40s + 16t = 0$ **03**
 (ii) Solve $p^2 + q^2 = x + y$ **04**

OR

- Q.4** (a) Solve $(mz - ny)p + (nx - lz)q = ly - mx$ **03**

- (b) Form a partial differential equation by eliminating the arbitrary functions from $f(x^2 - y^2, xyz) = 0$ 04
- (c) (i) Solve $(D^2 - DD' + D' - 1)z = \cos(x + 2y)$ 03
(ii) Solve using Charpit's Method $z^2 = pqxy$ 04
- Q.5** (a) Solve $(D^2 + 10DD' + 25D'^2)z = e^{3x+2y}$ 03
- (b) Solve $x \frac{\partial u}{\partial x} - 2y \frac{\partial u}{\partial y} = 0$ using method of separation of variables. 04
- (c) (i) Solve $(D^2 - D'^2)z = x - y$ 03
(ii) Solve $(2D^2 - 5DD' + 2D'^2)z = \sin(2x + y)$ 04
- OR**
- Q.5** (a) Solve $(1 - x)p + (z - y)q = 3 - z$ 03
- (b) Solve $2 \frac{\partial u}{\partial x} = \frac{\partial u}{\partial t} + u$ using method of separation of variables subject to the condition $u(x, 0) = 4e^{-3x}$ 04
- (c) Find the solution of the wave equation $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$ such that $y = a \cos pt$ when $x = l$ and $y = 0$ when $x = 0$ 07

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-III (NEW) EXAMINATION – WINTER 2023****Subject Code:3130005****Date:12-01-2024****Subject Name:Complex Variables and Partial Differential Equations****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Express $\left(\frac{2+i}{3-i}\right)^2$ into Polar form.	03
	(b) Find and plot the fourth roots of (-1) .	04
	(c) Solve $(D^2 + DD' + D' - 1)z = \sin(x + 2y)$	07
Q.2	(a) Determine a and b such that $u = ax^3 + bxy$ is harmonic.	03
	(b) Discuss the continuity of $f(z)$ at the origin.	04
	$f(z) = \frac{\bar{z}}{z}, \text{ if } z \neq 0$ $= 0, \text{ if } z = 0$	
	(c) Show that the function $u = e^x \cos y$ is harmonic. Find the conjugate function v and express $u + iv$ as an analytic function of z .	07
	OR	
	(c) Find the bilinear transformation which maps the points $1, -1, \infty$ onto the points $1 + i, 1 - i, 1$ respectively. Also, find its fixed points.	07
Q.3	(a) Find the radius of convergence of the power series $\sum_{n=1}^{\infty} \frac{z^n}{2^{n+1}}$.	03
	(b) Separate $(\sqrt{i})^{\sqrt{i}}$ into real and imaginary parts.	04
	(c) State Cauchy's Integral Theorem and use it to find $\int_C \frac{e^{2z}}{z^2+1} dz$, where C is $ z = \frac{1}{2}$.	07
	OR	
Q.3	(a) Evaluate $\int_0^{2+i} z^2 dz$ along the line $y = \frac{x}{2}$.	03
	(b) Expand $f(z) = \cos z$ as a Taylor series about $z = 0$.	04
	(c) Write Cauchy's Integral formula and hence evaluate: $\int_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz$, where C is $ z = 3$.	07
Q.4	(a) Classify the singular point $z = 0$ for function $f(z) = \frac{1}{z^4 - 4z^2}$.	03
	(b) Find the complete integral of $p - 3x^2 = q^2 - y$	04
	(c) Obtain the Laurent's series for the function $f(z) = \frac{1}{z(1-z)}$ in the regions : (i) $ z + 1 < 1$ (ii) $1 < z + 1 < 2$ (iii) $ z + 1 > 2$	07
	OR	
Q.4	(a) Derive partial differential equation by eliminating a and b from $z = (x - a)^2 + (y - b)^2$.	03

- (b) Using Cauchy's residue theorem, evaluate $\int_c \frac{5z-2}{z(z-1)} dz$; $|z| = 2$. 04
- (c) Solve $x(y-z)p + y(z-x)q = z(x-y)$. 07
- Q.5** (a) Evaluate $\int_0^\infty \frac{1}{1+x^2} dx$ using contour integration. 03
- (b) Solve $(D^2 - 4DD' + 4D'^2)z = e^{2x+3y}$. 04
- (c) Find the solution of the wave equation $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$ such that $y = a \cos pt$ when $x = l$, and $y = 0$ when $x = 0$. 07
- OR**
- Q.5** (a) Solve $(2D^2 + 5DD' + 2D'^2)z = 0$. 03
- (b) Solve $p(1+q) = qz$. 04
- (c) Using the method of separation of variable, find the solution of $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$, $u(x, 0) = 6e^{-3x}$. 07

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-III (NEW) EXAMINATION – SUMMER 2024****Subject Code:3130005****Date:16-07-2024****Subject Name: Complex Variables and Partial Differential Equations****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		MARKS
Q.1	(a) Represent $\frac{(1+i)^2}{1-i}$ in a+ib or u+iv form and find its modulus and argument	03
	(b) Define a harmonic function. Show that $u(x, y) = x^2 - y^2$ is harmonic and find the corresponding analytic function $f(z) = u(x, y) + iv(x, y)$	04
	(c) Find the bilinear transformation which maps the points $z=i, 1, -i$ onto the points $w=-i, 1, i$ respectively.	07
Q.2	(a) Evaluate $\int_0^{4+2i} \bar{z} dz$ along the curve $z = t^2 + it$	03
	(b) Find the centre and radius of convergence of the given power series. $\sum_{n=0}^{\infty} \frac{(2n)!}{(n!)^2} (z - 3i)^n$	04
	(c) Define analytic function and If $f(z)$ is analytic and $ f(z) = c$ then show that $f(z)$ is constant	07
	OR	
	(c) Evaluate $\oint_C \frac{\cos \pi z^2}{(z-1)(z-2)} dz$; C is $ z = 3$	07
Q.3	(a) Define: Singular point, Isolated singular point, Residue and Explain types of isolated singular points.	03
	(b) Find the Laurent series of $\frac{7z-2}{(z+1)z(z-2)}$; $1 < z+1 < 3$.	04
	(c) Evaluate using Cauchy residue theorem $\oint_C \frac{2z+6}{z^2+4} dz$ where $C: z-i = 2$	07
	OR	
Q.3	(a) Find the pole and its order of following functions 1. $f(z) = \frac{\sin z}{z^4}$ 2. $f(z) = \frac{1}{(z-5)^3(z^2-4)}$	03
	(b) Find the residues at singular points of $f(z) = \frac{z^2}{(z-1)^2(z+2)}$	04

- (c) Evaluate following real integration using residue theorem. **07**

$$\int_0^{2\pi} \frac{1}{(2 + \cos\theta)^2} d\theta$$

- Q.4 (a)** Form PDE by eliminating arbitrary Functions $F(x + y + z, x^2 + y^2 - z^2) = 0$ **03**

- (b) Solve following Linear Partial Differential Equations **04**

1) $xp + yq = x - y$

2) $(z - y)p + (x - z)q = (y - x)$

- (c) Solve following Non- linear partial differential equations using Charpit's method $px + qy = pq$ **07**

OR

- Q.4 (a)** Find the order of the following PDE (1 to 3). And check whether the equations are linear, quasilinear or nonlinear. **03**

1) $\frac{\partial u}{\partial x} - \frac{\partial u}{\partial t} = 0$, 2) $\frac{\partial^2 u}{\partial x^2} - e^{2x} \frac{\partial^2 u}{\partial t^2} = u^3$, 3) $u_y u_{yy} + (u_x)^2 = 0$

- (b) Solve following Non- linear partial differential equations. **04**

1) $p^2 - q^2 = x - y$

2) $p(1 + q) = qz$

- (c) Solve following Partial Differential Equations **07**

1) $(x^2 - y^2 - z^2)p + 2xyq = 2xz$

2) $\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial^2 z}{\partial x \partial y} - 8 \frac{\partial^2 z}{\partial y^2} = 0$

- Q.5 (a)** Classify second order homogeneous partial differential equations as elliptic, parabolic or hyperbolic **03**

1) $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial x \partial y} - 12 \frac{\partial^2 z}{\partial y^2} = 0$ 2) $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial x \partial y} = 0$

- (b) Using method of separation of variables solve $\frac{\partial u}{\partial t} + u = 2 \frac{\partial u}{\partial x}$ **04**
given that $u(x, 0) = 4e^{-3x}$

- (c) Determine solution of two-dimensional Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ **07**
which satisfy the condition $u(0, y) = u(l, y) = u(x, 0) = u(x, b) = \sin \frac{n\pi x}{l}$

OR

- Q.5 (a)** Solve partial differential equations by direct integration method **03**

$\frac{\partial^2 z}{\partial x^2} = \sin x$

- (b) $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial x \partial y} - 12 \frac{\partial^2 z}{\partial y^2} = 3e^{2x-3y}$ **04**
Solve second order homogeneous PDEs

- (c) Find the solution of One- Dimensional Wave Equation. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-III (NEW) EXAMINATION – WINTER 2024****Subject Code:3130005****Date:21-11-2024****Subject Name: Complex Variables and Partial Differential Equations****Time: 10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		MARKS
Q.1	(a) Find the principal argument $Arg(z)$ when $z = \frac{-2}{1+i\sqrt{3}}$.	03
	(b) Find all the values of following: $(1-i)^{\frac{2}{3}}$.	04
	(c) Find the bilinear transformation which maps the points $z = 0, -i, -1$ into $w = i, 1, 0$ respectively.	07
Q.2	(a) Show that $f(z) = z^3$ is analytic everywhere.	03
	(b) Evaluate $\int (x^2 - iy^2) dz$ along the parabola $y = 2x^2$ from $(1,2)$ to $(2,8)$.	04
	(c) Find Laurent's series expansion in powers of z that represent $f(z) = \frac{1}{z^2(1-z)}$ for the following domains: (i) $ z < 1$ (ii) $ z > 1$	07
	OR	
	(c) Find the image of the half-plane $x > c$, when $c > 0$ under the transformation $w = \frac{1}{z}$. Show the regions graphically.	07
Q.3	(a) If $u + iv$ is analytic, show that $v - iu$ and $-v + iu$ are also analytic.	03
	(b) Show that $u(x, y) = x^2 - y^2 + x$ is harmonic. Find the corresponding analytic function $f(z) = u + iv$.	04
	(c) Evaluate $\int_C \frac{\cos \pi z}{z^2 - 1} dz$, where C is the rectangle whose vertices are $2 \pm i, -2 \pm i$.	07
	OR	
Q.3	(a) Obtain the residue of $f(z) = \frac{z-3}{(z+1)(z+2)}$ at it's poles.	03
	(b) Evaluate $\int_C \frac{e^{2z}}{(z+1)^4} dz$, where C is the circle $ z = 2$.	04
	(c) Evaluate $\int_0^\infty \frac{x \sin x}{x^2 + 9} dx$ using residue.	07
Q.4	(a) Form a partial differential equation for the equation $(x-a)(y-b)-z^2 = x^2 + y^2$	03
	(b) Solve: $xp + yq = 3z$	04

- (c) Solve by Charpit's Method: $p = (z + qy)^2$ 07
- OR**
- Q.4** (a) Form a partial differential equation by eliminating the arbitrary function from $z = f(x^2 - y^2)$ 03
- (b) Solve $x(y - z)p + y(z - x)q = z(x - y)$ 04
- (c) Solve $(D^2 - 2DD' + D'^2)z = e^{x+2y} + x^3$ 07
- Q.5** (a) Solve: $p - x^2 = q + y^2$ 03
- (b) Solve $(D^2 - 2DD' + D'^2)z = \tan(x + y)$ 04
- (c) Solve $(D^2 + DD' - 6D'^2)z = \sin(2x + y)$ 07
- OR**
- Q.5** (a) Solve: $(p - q)(z - px - qy) = 1$ 03
- (b) Solve: $\frac{\partial^3 z}{\partial x^3} - 3\frac{\partial^3 z}{\partial x^2 \partial y} + 2\frac{\partial^3 z}{\partial y^3} = 0$ 04
- (c) Using the method of separation of variables, solve $\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u$, $u(x, 0) = 6e^{-3x}$ 07

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-III EXAMINATION – SUMMER 2025****Subject Code:3130005****Date:13-06-2025****Subject Name:Complex Variables and Partial Differential Equations****Time:02:30 PM TO 05:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	Marks
Q.1 (a) Find real and Imaginary parts of $z = \frac{3i}{2+3i}$.	3
(b) Find z , if $Arg(z+1) = \frac{\pi}{6}$ and $Arg(z-1) = \frac{2\pi}{3}$.	4
(c) Determine an analytic function whose real part is $u(x,y) = e^{2x}(x \cos 2y - y \sin 2y)$.	7
Q.2 (a) Evaluate $\lim_{z \rightarrow i} \frac{z-i}{z^2+1}$.	3
(b) Find $\log(1-i)$ and $\log(1+i)$.	4
(c) Find the bilinear transformation that maps $z_1 = 0, z_2 = 1, z_3 = \infty$ onto $w_1 = -1, w_2 = -i, w_3 = 1$ respectively.	7
OR	
(c) Verify that $u = 2x - x^3 + 3xy^2$ is harmonic in the whole complex plane and finds it's harmonic conjugate function $v(x,y)$.	7
Q.3 (a) Evaluate $\int_C \frac{e^z}{(z-1)(z-3)} dz$ counter clockwise around the circle $ z = 2$.	3
(b) Determine the residues of the function $f(z) = \frac{z+1}{(z^2-16)(z+2)}$ at each of its poles.	4
(c) Evaluate $\int_C Re(z^2) dz$, where C is the boundary of the square with vertices $0, i, 1+i, 1$ in the clockwise wise direction.	7
OR	
Q.3 (a) Classify the poles of $f(z) = \frac{1}{z^2-z^6}$.	3
(b) Find the Laurent's series of the function $f(z) = \frac{1}{(z+2)(z+4)}$ for the region $2 < z < 4$.	4
(c) Using Residue theorem evaluate the integral $\int_0^\pi \frac{1}{17-8 \cos \theta} d\theta$.	7
Q.4 (a) Determine the residues of the function $f(z) = \frac{3z-4}{z(z-1)(z-2)}$ at each of its poles.	3
(b) Solve $\frac{\partial^3 z}{\partial x^2 \partial y} = \sin(3x + 2y)$.	4
(c) Solve $x^2(y-z)p + y^2(z-x)q = z^2(x-y)$	7

OR

- Q.4** (a) Find the fixed points of $w = 1 + \frac{1}{z}$. 3
- (b) Solve $\frac{\partial^2 z}{\partial x^2} + 4 \frac{\partial^2 z}{\partial x \partial y} - 5 \frac{\partial^2 z}{\partial y^2} = \sin(2x + 3y)$. 4
- (c) Solve: $\frac{\partial^2 z}{\partial x \partial y} = \sin x \sin y$, for which $\frac{\partial z}{\partial y} = -2 \sin y$ when $x = 0$ and $z = 0$ when y is an odd multiple of $\frac{\pi}{2}$. 7

- Q.5** (a) Form the partial differential equation by eliminating arbitrary function from $z = f(x + at) + g(x - at)$. 3
- (b) Solve $p^2 + q^2 = x + y$. 4
- (c) Using method of separation of variables solve $3 \frac{\partial u}{\partial x} + 2 \frac{\partial u}{\partial y} = 0$, given that $u(x, 0) = 4e^{-x}$. 7

OR

- Q.5** (a) Solve $z = px + qy + \sqrt{1 + p^2 + q^2}$. 3
- (b) Solve $\frac{y^2 z}{x} p + xzq = y^2$. 4
- (c) The base of a semi-infinite strip of a metal plate is 30 cm and it kept at 100°C . The two long edges are at zero temperature. Find the temperature at any point 15 cm away from the base and situated midway between the long edges. 7

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER- IV EXAMINATION – SUMMER 2020****Subject Code: 3140610****Date: 02/11/2020****Subject Name: Complex Variables and Partial Differential Equations****Time: 10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

		Mark s
Q.1	(a) Show that the function $u = x^2 - y^2$ is harmonic and find the corresponding analytic function.	03
	(b) Find the fourth roots of -1 .	04
	(c) (i) Find the image of the infinite strip $0 \leq x \leq 1$ under the transformation $w = iz + 1$. Sketch the region.	03
	(ii) Write the function $f(z) = z + \frac{1}{z}$ in the form $f(z) = u(r, \theta) + iv(r, \theta)$.	04
Q.2	(a) Evaluate $\int_C (x^2 + ixy) dz$ from $(1,1)$ to $(2,4)$ along the curve $x = t, y = t^2$.	03
	(b) Find the bilinear transformation which transforms $z = 2, 1, 0$ into $w = 1, 0, i$	04
	(c) (i) Evaluate $\oint_C \frac{dz}{z^2 + 1}$, where C is $ z + i = 1$, counter clockwise.	03
	(ii) Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!} z^n$.	04
	OR	
	(c) (i) Find the roots of the equation $z^2 + 2iz + (2 - 4i) = 0$.	03
	(ii) Find the roots of $\log z = i \frac{\pi}{2}$.	04
Q.3	(a) Find $\oint_C \left(\frac{3}{z-i} - \frac{6}{(z-i)^2} \right) dz$, where $C : z = 2$.	03
	(b) Find the residues of $f(z) = \frac{1}{(z-1)^2(z-3)}$, has a pole at $z = 3$ and a pole of order 2 at $z = 1$.	04
	(c) Expand $f(z) = \frac{1}{(z+1)(z+3)}$ in a Laurent series for the regions (i) $ z < 1$, (ii) $1 < z < 3$, (iii) $ z > 3$.	07
	OR	
Q.3	(a) Evaluate $\oint_C \frac{e^z}{z+i} dz$, where $C : z-1 = 1$.	03
	(b) Evaluate by using Cauchy's residue theorem $\int_C \frac{5z-2}{z(z-1)} dz ; z = 2$.	04

- (c) Expand $f(z) = \frac{1}{(z+1)(z-2)}$ in Laurent's series for the regions (i) $|z| < 1$, **07**
(ii) $1 < |z| < 2$, (iii) $|z| > 2$.
- Q.4** (a) Solve $yq - xp = z$. **03**
(b) Derive partial differential equation by eliminating the arbitrary constants a and b from $z = (x^2 + a)(y^2 + b)$. **04**
(c) (i) Solve the p.d.e. $r - 3as + 2a^2t = 0$. **03**
(ii) Find the complete integral of $p(1+q) = qz$. **04**
- OR**
- Q.4** (a) Find the solution of $(y-z)p + (z-x)q = x-y$. **03**
(b) Form the partial differential equation by eliminating the arbitrary function from $\phi(x+y+z, x^2+y^2+z^2) = 0$. **04**
(c) (i) Solve the p.d.e. $s + p - q = z + xy$. **03**
(ii) Solve by Charpit's method $q = 3p^2$. **04**
- Q.5** (a) Solve $(D^2 + 2DD' + D'^2)z = e^{2x+3y}$ **03**
(b) Solve the p.d.e. $u_x = 4u_y, u(0, y) = 8e^{-3y}$ using the method of separation of variables. **04**
(c) Find the solution of the wave equation $u_{tt} = c^2 u_{xx}$, $0 \leq x \leq L$ with the conditions $u(0, t) = u(L, t) = 0; t > 0$, $u(x, 0) = \frac{\pi x}{L}, u_t(x, 0) = 0; 0 \leq x \leq L$. **07**
- OR**
- Q.5** (a) Solve the p.d.e. $r + s + q - z = 0$. **03**
(b) Solve $xu_x - 2yu_y = 0$ using the method of separation of variables. **04**
(c) Find the solution of $u_t = c^2 u_{xx}$ together with the initial and boundary conditions $u(0, t) = u(l, t) = 0; t \geq 0$ and $u(x, 0) = \sin \frac{\pi x}{l}; 0 \leq x \leq l$. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-IV (NEW) EXAMINATION – WINTER 2020****Subject Code:3140610****Date:17/02/2021****Subject Name:Complex Variables and Partial Differential Equations****Time:02:30 PM TO 04:30 PM****Total Marks:56****Instructions:**

1. Attempt any FOUR questions out of EIGHT questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

		Marks
Q.1	(a) Show that the function $u = x^3 - 3xy^2$ is harmonic.	03
	(b) Find the fifth root of unity.	04
	(c) (i) Determine and sketch the image of $ z = 1$ under the transformation $w = z + i$.	03
	(ii) Find the real and imaginary parts of $f(z) = \frac{3i}{2+3i}$.	04
Q.2	(a) Evaluate $\int_C (x^2 - iy^2) dz$ along the parabola $y = 2x^2$ from (1,2) to (2,8)	03
	(b) Find the bilinear transformation that maps the points $z = \infty, i, 0$ into the points $w = 0, i, \infty$ respectively.	04
	(c) (i) Evaluate $\oint_C \frac{\sin 3z}{z + \pi/2} dz$ where C is the circle $ z = 5$.	03
	(ii) For which values of z does the series $\sum_{n=0}^{\infty} n! z^n$ convergent?	04
Q.3	(a) Evaluate $\oint_C \frac{e^{-z}}{z+1} dz$ where C is the circle $ z = 1/2$.	03
	(b) Find the residue $\text{Res}(f(z), -1)$ for $f(z) = \frac{z^2 - 2z}{(z+1)^2(z^2+4)}$.	04
	(c) Expand $f(z) = \frac{1}{(z+2)(z+4)}$ in the region (i) $ z < 2$, (ii) $2 < z < 4$, (iii) $ z > 4$.	07
Q.4	(a) Evaluate $\oint_C \frac{2z+6}{z^2+4} dz$ where C is $ z-i = 2$.	03
	(b) Using Cauchy's Residue Theorem evaluate $\int_C \frac{e^{2z}}{(z+1)^3} dz$, where C is the ellipse $4x^2 + 9y^2 = 16$.	04

- (c) Expand $\frac{1}{z(z^2 - 3z + 2)}$ about $z = 0$ for the region (i) $0 < |z| < 1$, **07**
(ii) $1 < |z| < 2$, (iii) $|z| < 2$.
- Q.5** (a) Solve $y^2 p - xyq = x(z - 2y)$. **03**
(b) Derive p.d.e. by eliminating a and b from $z = (x - a)^2 + (y - b)^2$. **04**
(c) (i) Solve $(D^3 - 3D^2 D' + 2D'^3)z = 0$. **03**
(ii) Find the complete integral of $p^2 = qz$. **04**
- Q.6** (a) Solve $x^2 p + y^2 q = z^2$. **03**
(b) Form a p.d.e. by eliminating the arbitrary function from $z = f(x^2 - y^2)$. **04**
(c) (i) Solve $(D - D' - 1)(D - D' - 2)z = e^{2x-y} + x$. **03**
(ii) Solve $(p^2 + q^2)y = qz$ by Charpit's method. **04**
- Q.7** (a) Solve $2r - 5s + 2t = 24(y - x)$. **03**
(b) Solve the p.d.e. $u_x + u_y = 2(x + y)u$. **04**
(c) A tightly stretched string with fixed end points $x = 0$, $x = l$ is initially at rest in its equilibrium position. If it is set vibrating giving each point a velocity $\lambda x(l - x)$, find the displacement $u(x, t)$. **07**
- Q.8** (a) Solve $(D^2 - D'^2 + D - D')z = 0$. **03**
(b) Solve the p.d.e. $u_{xx} = 16u_y$. **04**
(c) Find the solution of one dimensional heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ **07**
satisfying the conditions $u(0, t) = u(\pi, t) = 0$ for $t > 0$ and $u(x, 0) = \pi - x, 0 < x < \pi$.

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-IV (NEW) EXAMINATION – SUMMER 2021****Subject Code:3140610****Date:07/09/2021****Subject Name:Complex Variables and Partial Differential Equations****Time:02:30 PM TO 05:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Find the real and imaginary parts of $f(z) = iz + 2\bar{z}$.	03
	(b) State De-Moivre's formula and evaluate $(1 + i)^{96} + (1 - i)^{96}$.	04
	(c) Define analytic function. Show that $u(x, y) = x^2 - y^2 - y$ is harmonic function, find a harmonic conjugate $v(x, y)$.	07
Q.2	(a) Find the image of the region $ z < 2$ under the transformation $w = 3z + i$.	03
	(b) Find all solution of $\sin z = 2$.	04
	(c) Expand $f(z) = \frac{1}{(z+4)(z+2)}$ valid for the region	07
	(i) $ z < 2$ (ii) $2 < z < 4$ (iii) $ z > 4$.	
	OR	
	(c) Determine the Mobius transformation which maps $z_1 = -1, z_2 = i, z_3 = 1$ into $w_1 = 0, w_2 = i, w_3 = \infty$. Hence, find the image of $ z < 1$.	07
Q.3	(a) Check whether the function $f(z) = x^2 + y^2 - i2xy$ is analytic or not.	03
	(b) Evaluate $\oint_C \frac{e^z}{z(z-1)} dz$ around the circle $C: z = 2$.	04
	(c) Evaluate the followings:	07
	(i) $\int_C \frac{\cos \pi z^2}{(z-2)(z-1)} dz$, counter clockwise around the circle $C: z =3$.	
	(ii) $\int_C \frac{dz}{z^2+4}$, where C is the unit circle.	
	OR	
Q.3	(a) Show that the function $f(z) = \bar{z}$ is nowhere differentiable.	03
	(b) Expand $f(z) = z^2 \exp\left(\frac{1}{z}\right)$ in Laurent's series about $z = 0$ and classify the singularity.	04
	(c) Using residue theorem, evaluate $\int_0^{2\pi} \frac{d\theta}{5-3\sin\theta}$.	07
Q.4	(a) Find the value of $\int_0^{1+i} (\bar{z})^2 dz$ along the line $y = x$.	03
	(b) Solve the partial differential equation $(D^2 - 3DD' + 2D'^2)z = \sin(x + 2y)$.	04
	(c) Obtain the complete integral of the followings:	07
	(i) $q - p + x - y = 0$.	
	(ii) $q^2 = z^2 p^2 (1 - p^2)$.	
	OR	
Q.4	(a) Classify the Partial differential equation $\frac{\partial^2 u}{\partial x^2} + 7 \frac{\partial^2 u}{\partial x \partial y} + 9 \frac{\partial^2 u}{\partial y^2} = 0$.	03

- (b) Solve $(x^2 - yz)p + (y^2 - xz)q = z^2 - xy$. 04
- (c) Find the general solution of the partial differential equation $\frac{\partial^2 u}{\partial x^2} = 25 \frac{\partial^2 u}{\partial y^2}$ by method of separation of variables. 07
- Q.5** (a) Solve $px + qy = 3z$. 03
- (b) Using Charpit's method, solve $q = 3p^2$. 04
- (c) A tightly stretched string with fixed end points at $x = 0$ and $x = 20$ is initially given by the deflection $f(x) = kx(20 - x)$. If it is released from this position, then find the deflection of the string. 07

OR

- Q.5** (a) Solve $2 \frac{\partial^2 z}{\partial x^2} + 5 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} = 0$. 03
- (b) Solve $\frac{\partial^2 z}{\partial x^2} + z = 0$, given that when $x = 0, z = e^y$ and $\frac{\partial z}{\partial x} = y$. 04
- (c) A rod of 30 cm long has its ends A and B are kept at 20°C and 80°C respectively until steady state conditions prevail. The temperature at each end is suddenly reduced to 0°C and kept so. Find the resulting temperature $u(x, t)$ from the end A . 07

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-IV (NEW) EXAMINATION – WINTER 2021****Subject Code:3140610****Date:24/12/2021****Subject Name:Complex Variables and Partial Differential Equations****Time:10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		MARKS
Q.1	(a) Prove that $i^i = e^{-(4n+1)\frac{\pi}{2}}$	03
	(b) Show that the function $u=x^2-y^2$ is harmonic and find the corresponding analytic function	04
	(c) State DE MOIVRE'S Theorem and Find the fifth root of unity	07
Q.2	(a) Prove that $\sinh^{-1} x = \log(x + \sqrt{x^2 + 1})$	03
	(b) Show that $f(z) = \frac{\operatorname{Re}(z^2)}{ z }, z \neq 0$ $= 0, z=0$ Is continuous at $z=0$.	04
	(c) State Necessary condition for function to be analytic and Show that neither $f(z)=\bar{z}$ nor $f(z)= z $ is an analytic function.	07
	OR	
	(c) Using Residue Evaluate $\int_0^\infty \frac{x^2 dx}{(x^2+1)(x^2+4)}$.	07
Q.3	(a) Evaluate $\int_c (x^2 - iy^2) dz$ along the parabola $y=2x^2$ from (1,2) to (2,8)	03
	(b) Discuss the convergence of the series. $\sum_{n=0}^\infty \frac{(2n)!(z-3i)^n}{(n!)^2}$	04
	(c) State Cauchy Integral Formula and evaluate $\int_c \frac{z-1}{(z+1)^2(z-2)} dz$, where c is the circle $ z-i =2$	07
	OR	
Q.3	(a) Expand $f(z)=\frac{z-\sin z}{z^2}$ at $z=0$, classify the singular point $z=0$.	03
	(b) Define Mobius Transformation which maps $z=0, i, 1$ into $w=i, -1, \infty$	04
	(c) Determine the poles of the function $f(z)=\frac{z^2}{(z-1)^2(z+2)}$ and residue at each pole. Hence evaluate $\oint_c f(z)$ where c is the circle $ z =3$.	07
Q.4	(a) Solve $z^2(p^2 z^2 + q^2) = 1$	03

- (b) Solve $x(y-z)p+y(z-x)q=z(x-y)$ **04**
 (c) Using method of separation of variable solve $\frac{\partial u}{\partial x}=2\frac{\partial u}{\partial t}+u$, **07**
 where $u(x,0)=6e^{-3x}$

OR

- Q.4** (a) Solve $q=3p^2$ by Charpit method **03**
 (b) Solve $(D^2+10DD'+25D'^2)Z=e^{3x+2y}$ **04**
 (c) A tightly stretched string with fixed end points $x=0$ and $x=l$ is initially in the equilibrium position. If it is set vibrating by giving to each of its points Q , a velocity of $v_0 \sin^3 \frac{\pi x}{l}$. Find the displacement $y(x,t)$. **07**

- Q.5** (a) Form a partial differential equation by eliminating arbitrary function $xyz=\phi(x+y+z)$ **03**
 (b) Classify the following partial differential equation **04**
 1) $2\frac{\partial^2 u}{\partial x^2}+4\frac{\partial^2 u}{\partial x \partial y}+3\frac{\partial^2 u}{\partial y^2}=6$
 2) $\frac{\partial^2 u}{\partial x^2}+4\frac{\partial^2 u}{\partial x \partial y}+4\frac{\partial^2 u}{\partial y^2}=0$

- (c) Find the solution of wave equation $\frac{\partial^2 u}{\partial t^2}=c^2 \frac{\partial^2 u}{\partial x^2}$, **07**
 $0 \leq x \leq L$ satisfying the condition $u(0,t)=u(L,t)=0$
 $u_t(x,0)=0, u(x,0)=\frac{\pi x}{L}, 0 \leq x \leq L$

OR

- Q.5** (a) Solve $\frac{\partial^2 z}{\partial x^2}=z$ **03**
 (b) Find the solution of $u_t=c^2 u_{xx}$ together with the initial and boundary conditions $u(0,t)=u(l,t)=0$ for all $t \geq 0$ and $u(x,0)=\sin \frac{\pi x}{l}, 0 \leq x \leq l$ **04**
 (c) A rod 30 cm long has its end A and B Kept at 20° and 80° **07**
 c respectively until steady state condition prevail. The temperature at each end is then suddenly reduced to 0° kept so. Find the resulting temperature function $u(x,t)$ from the end A.

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-IV (NEW) EXAMINATION – SUMMER 2022****Subject Code:3140610****Date:02-07-2022****Subject Name:Complex Variables and Partial Differential Equations****Time:10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	MARKS
Q.1 (a) Find an analytic function $f(z) = u + iv$ if $u = x^3 - 3xy$.	03
(b) Find the fourth roots of -1.	04
(c) (i) Find the image of infinite strip $0 \leq x \leq 1$ under the transformation $w = iz + 1$.	03
(ii) Separate real and imaginary parts of $f(z) = z^2$.	04
Q.2 (a) Evaluate $\int_C (x^2 + ixy) dz$ from (1, 1) to (2, 4) along the curve $x = t, y = t^2$.	03
(b) Determine the mobius transformation that maps $z_1 = 0, z_2 = 1, z_3 = \infty$ onto $w_1 = -1, w_2 = -i, w_3 = 1$ respectively.	04
(c) (i) Evaluate $\oint_C \frac{e^z}{z(1-z)^3} dz$, where C is $ z = \frac{1}{2}$.	03
(ii) Find the radii of convergence of $\sum_{n=1}^{\infty} \frac{z^n}{2^n + 1}$.	04
OR	
(c) Find the image of $ z - 1 = 1$ under the mapping $w = \frac{1}{z}$.	07
Q.3 (a) Evaluate $\oint_C \frac{e^{2z}}{(z+1)^4} dz$, where C is the circle $ z = 2$.	03
(b) Find $\operatorname{Re} s(f(z), 4i)$, where $f(z) = \frac{z}{z^2 + 16}$.	04
(c) Expand $f(z) = \frac{1}{(z-1)(z+2)}$ in Laurent's series in the region	07
(i) $ z < 1$, (ii) $1 < z < 2$, (iii) $ z > 2$.	
OR	
Q.3 (a) Evaluate $\oint_C (x^2 - y^2 + 2ixy) dz$, where C is the circle $ z = 1$.	03
(b) Evaluate $P.V. \int_{-\infty}^{\infty} \frac{x \cos x}{x^2 + 9} dx$.	04
(c) Find Laurent's series that represent $f(z) = \frac{1}{z(z-1)}$ in the region	07
(i) $0 < z < 1$, (ii) $0 < z-1 < 1$.	

- Q.4 (a)** Solve $\frac{y-z}{yz}p + \frac{z-x}{zx}q = \frac{x-y}{xy}$. **03**
- (b)** Derive partial differential equation by eliminating arbitrary constants a and b from $z = (x+a)(y+b)$. **04**
- (c)** (i) Solve $\frac{\partial^3 z}{\partial x^3} = 0$. **03**
- (ii) Find complete integral of $p^2 + q^2 = z$. **04**
- OR**
- Q.4 (a)** Solve $xp + yq = x - y$. **03**
- (b)** Form a partial differential equation by eliminating arbitrary function from $z = f(x/y)$. **04**
- (c)** (i) Solve $(D^2 - D'^2 + D - D')z = 0$. **03**
- (ii) Solve $q = 3p^2$ by Charpit's method. **04**
- Q.5 (a)** Solve $(r + 3s + 2t) = x + y$ **03**
- (b)** Solve the p.d.e. $u_{xy} = -u_x$. **04**
- (c)** Find the deflection $u(x, t)$ of the vibrating string of length π and ends fixed, corresponding to zero velocity and initial deflection $f(x) = k(\sin x - \sin 2x)$. **07**
- OR**
- Q.5 (a)** Solve $(D^2 + DD' + D' - 1)z = \sin(x + 2y)$. **03**
- (b)** Solve the p.d.e. $u_x + u_y = 2(x + y)u$. **04**
- (c)** Find the solution of $u_t = c^2 u_{xx}$ together with the initial and boundary conditions $u(0, t) = u(l, t) = 0$ for all $t \geq 0$ and $u(x, 0) = \sin \frac{\pi x}{l}$, $0 \leq x \leq l$. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-IV(NEW) EXAMINATION – WINTER 2022****Subject Code:3140610****Date:16-12-2022****Subject Name:Complex Variables and Partial Differential Equations****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	Marks
Q.1 (a) Find the real and imaginary parts of $f(z) = z^2 + \bar{z}$.	03
(b) Evaluate $(1 + i\sqrt{3})^{60} + (1 - i\sqrt{3})^{60}$.	04
(c) Define Harmonic function. Show that $u(x, y) = \sin z$ is analytic everywhere. Also, find $f'(z)$.	07
Q.2 (a) Find the image of the region $ z > 2$ under the transformation $w = 4z$	03
(b) Find all solution of $e^z = 1 + i$.	04
(c) Expand $f(z) = \frac{1}{(z-2)(z-3)}$ valid for the region (i) $ z < 2$ (ii) $2 < z < 3$ (iii) $ z > 3$.	07
OR	
(c) Show that $u(x, y) = y^3 - 3x^2y$ is harmonic in some domain D and find the conjugate $v(x, y)$	07
Q.3 (a) Check whether the function $f(z) = xy + iy$ is analytic or not at any point.	03
(b) Evaluate $\oint_C \frac{\cos z}{(z-1)(z-2)} dz$ around the circle $C: z = 3$.	04
(c) Evaluate the followings: (i) $\int_C \frac{2z+3}{z^2-4} dz$, counter clockwise around the circle $C: z-2 =1$. (ii) $\int_C \frac{e^z+z}{z^2-1} dz$, where $C: z =2$	07
OR	
Q.3 (a) Expand $f(z) = \frac{\cos z}{z^2}$ in Laurent's series about $z = 0$ and identify the singularity.	03
(b) Determine the bilinear transformation which maps the points $0, \infty, i$ into $\infty, 1, 0$	04
(c) Using residue theorem, evaluate $\int_0^{2\pi} \frac{4d\theta}{5+4\sin\theta}$.	07
Q.4 (a) Evaluate $\int_C Re(z)dz$, where c is the shortest path from $1+i$ to $3+2i$	03
(b) Solve the partial differential equation $(D^2 - 5DD' + 6D'^2)z = e^{x+y}$.	04
(c) Obtain the complete integral of the followings: (i) $p^2 - q^2 = x - y$. (ii) $z = px + qy - 2\sqrt{pq}$	07

OR

- Q.4** (a) Find the Laurent's series that represents the function $f(z) = z^2 \sin\left(\frac{1}{z^2}\right)$ in the domain $0 < |z| < \infty$ **03**
(b) Solve $x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2)$ **04**
(c) Find the general solution of the partial differential equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ by method of separation of variables. **07**

- Q.5** (a) Solve $pq = p + q$ **03**
(b) Find a complete integral of the equation $p^2y(1 + x^2) = qx^2$ **04**
(c) If a string of length l is initially at rest in equilibrium position and each of its points is given the velocity $\left(\frac{\partial y}{\partial t}\right)_{t=0} = k \sin^3\left(\frac{\pi x}{l}\right)$, x being the distance from an end point. Find the displacement of the string at any point. **07**

OR

- Q.5** (a) Solve $\frac{\partial^2 z}{\partial x^2} = \sin x$. **03**
(b) Solve: $(y + z)p + (z + x)q = x + y$ **04**
(c) A homogeneous rod of conducting material of length 100cm has its ends kept at zero temperature and the temperature initially is $u(x, 0) = x$, $0 \leq x \leq 100$. Find the temperature $u(x, 0)$ at any time. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER– IV(NEW) EXAMINATION – SUMMER 2023****Subject Code:3140610****Date:17-07-2023****Subject Name:Complex Variables and Partial Differential Equations****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Prove that i^i is real.	03
	(b) Simplify $\frac{(\cos 5\theta - i \sin 5\theta)^2 (\cos 7\theta + i \sin 7\theta)^{-3}}{(\cos 4\theta - i \sin 4\theta)^9 (\cos \theta + i \sin \theta)^5}$	04
	(c) Solve $(D^2 + DD' - 6D'^2)z = \sin(2x + y)$	07
Q.2	(a) Check whether the function $f(z) = e^{\bar{z}}$ is analytic or not at any point.	03
	(b) Find and plot all values of $(8i)^{\frac{1}{3}}$.	04
	(c) Show that the function $u = e^x \cos y$ is harmonic. Determine its harmonic conjugate $v(x, y)$ and the analytic function $f(z) = u + iv$.	07
	OR	
	(c) Determine the region in the w-plane into which the triangle bounded by the lines $x = 0$, $y = 0$ and $x + y = 1$ in the z-plane is mapped under the transformation $w = 4z$.	07
Q.3	(a) State: i) Cauchy-Goursat Theorem ii) Liouville Theorem.	03
	(b) Find an upper bound for the absolute value of $\oint_C \frac{e^z}{z+1} dz$, where C is the circle $ z = 4$.	04
	(c) Write Cauchy's Integral formula and hence evaluate $\oint_C \frac{z+1}{z^4+2iz^3} dz$, where C is the circle $ z = 1$.	07
	OR	
Q.3	(a) Evaluate $\oint_C \frac{1}{z} dz$, where C is the circle $x = \cos t$, $y = \sin t$, $0 \leq t \leq 2\pi$.	03
	(b) Find the values of x and y if $e^z = \sqrt{3} + i$.	04
	(c) Evaluate $\int_C \frac{z+1}{z^4-4z^3+4z^2} dz$, where C is the circle $ z - 2 - i = 2$.	07
Q.4	(a) Identify the type of singularities of $f(z) = \frac{\tan z}{z}$.	03
	(b) Form a partial differential equation by eliminating the arbitrary function from $z = f(x^2 - y^2)$.	04
	(c) Express $f(z) = \frac{1}{z(z-1)}$ in a Laurent series valid for the following annular domains.	07
	(a) $0 < z < 1$ (b) $1 < z $ (c) $0 < z - 1 < 1$.	

OR

- Q.4** (a) Find the complete integral of $pq = k$, where k is a constant. **03**
 (b) Evaluate by the Residue Theorem $\int_C \frac{1}{(z-1)^2(z-3)} dz$, where the contour C is the rectangle defined by $x = 0, x = 4, y = -1, y = 1$. **04**
 (c) Solve $(x^2 - y^2 - z^2)p + 2xyq = 2xz$. **07**
- Q.5** (a) Test for singularity of $\frac{1}{z^2+1}$ and hence find the corresponding residues. **03**
 (b) Solve $\frac{\partial^3 z}{\partial x^2 \partial y} = \cos(2x + 3y)$. **04**
 (c) Using the method of separation of variable, find the solution of $\frac{\partial u}{\partial x} = 4 \frac{\partial u}{\partial y}$, given that $u(0, y) = 8e^{-3y}$. **07**
- OR**
- Q.5** (a) Solve $(D^3 - 2D^2D')z = 2e^{2x}$. **03**
 (b) Solve $p^2 + q^2 = x + y$. **04**
 (c) Solve $(D^2 + DD' - 6D'^2)z = y \cos x$. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-IV (NEW) EXAMINATION – WINTER 2023****Subject Code:3140610****Date: 06-02-2024****Subject Name: Complex Variables and Partial Differential Equations****Time: 10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

- Q. 1**
- (a) 1. State De Moivre's theorem on complex numbers. 01
 2. Write the formula of $\sin h(x+y)$ 01
 3. Find the modulus of $z = \frac{1}{\sqrt{3} + i}$ 01
- (b) Prove that $\sinh^{-1} z = \log \left\{ z + \sqrt{z^2 + 1} \right\}$ 04
- (c) Find all the roots of $x^{12} - 1 = 0$ and identify the roots which are also roots of $x^4 + x^2 + 1 = 0$. 07
- Q. 2**
- (a) Define limit of a complex function. Evaluate $\lim_{z \rightarrow i} \frac{z - i}{z^2 + 1}$ 03
- (b) Show that $f(z) = z \operatorname{Im}(z)$ is differentiable only at $z = 0$ and $f'(0) = 0$. 04
- (c) State the necessary and sufficient conditions for the complex function $f(z) = u + iv$ to be analytic function. Find the analytic function $f(z) = u + iv$, if $u = x^3 - 3xy^2$. Also determine its imaginary part v . 07
- OR**
- (c) Define harmonic function. Show that $u(x, y) = 2x - x^3 + 3xy^2$ is harmonic. Find also its harmonic conjugate v , where $f(z) = u + iv$. 07
- Q. 3**
- (a) State the following on complex analysis: 03
 (i) Cauchy–Goursat theorem (ii) Cauchy's integral formula (iii) Liouville's theorem.
- (b) Evaluate $\int_c (x^2 - iy^2) dz$ along the parabola $y = 2x^2$ from the points $(-2, 8)$ to $(2, 8)$. 04
- (c) 1. Evaluate $\oint_c \frac{e^{2z}}{(z+1)^4} dz$ where c is the circle $|z| = 3$. 03
 2. Evaluate $\oint_c \frac{\cos \pi z^2}{(z-1)(z-2)} dz$, where c is the circle $|z| = 3$. 04
- OR**
- Q. 3** (a) Find the upper bound for $\int_c e^z dz$ where c is the line segment joining the points $(0, 0)$ to $(1, 2\sqrt{2})$. 03

- (b) Find Laurent's series expansion in the power of z that represents the function $f(z) = \frac{1}{z^2(1-z)}$ in the domain $|z| > 1$. 04
- (c) 1. Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{(2n)!}{(n!)^2} (z-3i)^n$ 03
2. Find residues at each pole for $f(z) = \frac{1}{(z-3)(z-1)^2}$ 04
- Q. 4** (a) Form the partial differential equation from $z = xy + f(x^2 + y^2)$ 03
- (b) Solve $p^2 - q^2 = x - y$ 04
- (c) Solve $(x^2 - y^2 - z^2)p + 2xyq = 2xz$ 07
- OR**
- Q. 4** (a) Form the partial differential equation from $(x-a)(y-b) - z^2 = x^2 + y^2$. 03
- (b) Solve $p(1+q) = qz$ 04
- (c) Apply Charpit's method to solve $px + qy = pq$ 07
- Q. 5** (a) Discuss the classification of the second order linear partial differential equation. 03
- (b) Solve the equation $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$ by the method of separation of variables given that $u(x,0) = 6e^{-3x}$ 04
- (c) Solve $\frac{\partial^3 z}{\partial x^3} - 3 \frac{\partial^3 z}{\partial x^2 \partial y} + 4 \frac{\partial^3 z}{\partial y^3} = e^{x+2y}$ 07
- OR**
- Q. 5** (a) Classify the partial differential equation $\frac{\partial^2 u}{\partial x^2} + 2 \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 0$ as parabolic, elliptic or hyperbolic? 03
- (b) Solve the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ by the method of separation of variables given that $u(x,0) = \sin \pi x$ 04
- (c) The vibration of an elastic string is governed by the partial differential equation $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$. 07
- The length of the string is π and the ends are fixed. The initial velocity is zero and the initial deflection is $u(x,0) = 2(\sin x + \sin 3x)$. Find the deflection $u(x,t)$ of the vibrating string for $t > 0$.

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-IV (NEW) EXAMINATION – SUMMER 2024****Subject Code:3140610****Date:01-07-2024****Subject Name: Complex Variables and Partial Differential Equations****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Find $ z $, $\text{Arg } z$ and $\arg z$ for $z = \frac{-i}{1-i}$	03
	(b) Define a bilinear transformation. Find the Mobius transformation which sends the points $z = -1, 0, 1$ of the z -plane onto the points $w = -1, -i, 1$ respectively in the w -plane.	04
	(c) Define an analytic function. Show that $u(x, y) = 2x - x^3 + 3xy^2$ is a harmonic function. Determine also its harmonic conjugate $v(x, y)$ and the corresponding analytic function $f(z) = u + iv$.	07
Q.2	(a) Evaluate $\int_C \frac{e^z}{(z-3)^2} dz$, where $C: z = 2$.	03
	(b) Evaluate $\int_C \frac{1}{z^3(z+4)} dz$, where $C: z = 2$.	04
	(c) Evaluate $\int_0^{2+i} (\bar{z})^2 dz$, along the shortest path joining the two points $z = 0$ and $z = 2 + i$.	07
	OR	
	(c) Find the radius of convergence of	
	(i) $\sum_{n=1}^{\infty} \frac{(2n)!}{(n!)^2} (z - 3i)^n$	04
	(ii) $\sum_{n=0}^{\infty} (6 + 8i)z^n$	03
Q.3	(a) Find the Maclaurin's series for $f(z) = \frac{1}{1-z}$ about $z = 0$.	03
	(b) Expand the function $f(z) = \frac{1}{4z-z^2}$ in Laurent's series in powers of z under the condition $0 < z < 4$.	04
	(c) State Cauchy's Residue theorem. Using it integrate $f(z) = \frac{1}{z^3-z^4}$ counter clockwise around the circle $ z = \frac{1}{2}$.	07
	OR	
Q.3	(a) Expand $f(z) = \frac{1}{1-z}$ in powers of z , where $ z > 1$.	03
	(b) Prove that $\int_{-\infty}^{\infty} \frac{\cos x}{(x^2+1)^2} dx = \frac{\pi}{e}$, by using complex integrals.	04
	(c) Using Cauchy's Residue theorem, evaluate $\int_C \frac{z+1}{z^2-2z} dz$ counter clockwise around the circle $ z = 3$.	07

- Q.4** (a) Form a partial differential equation by eliminating arbitrary constants from $(x - a)(y - b) = x^2 + y^2 + z^2$ **03**
- (b) Using Lagrange's method, solve the partial differential equation $y^2p - xyq = x(z - 2y)$. **04**
- (c) Using Charpit's method, solve the partial differential equation $px + qy = pq$. **07**
- OR**
- Q.4** (a) Form a partial differential equation by eliminating arbitrary function from $z = xy + f(x^2 + y^2)$ **03**
- (b) Solve the non-linear partial differential equation $p^2 + q^2 = npq$. **04**
- (c) Using Charpit's method, solve the partial differential equation $px + qy + p^2 + q^2 - z = 0$ **07**
- Q.5** (a) Classify the partial differential equation $u_{xx} + 2u_{xy} + u_{yy} = 0$, as hyperbolic, parabolic or elliptic. **03**
- (b) Solve the partial differential equation $\frac{\partial^3 z}{\partial x^3} - 2 \frac{\partial^3 z}{\partial x^2 \partial y} = 2e^{2x}$ **04**
- (c) Find the solution of the one dimensional heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ satisfying the conditions $u(0, t) = u(\pi, t) = 0$; for $t > 0$ and $u(x, 0) = \pi - x$; for $0 < x < \pi$. **07**
- OR**
- Q.5** (a) Solve the partial differential equation $\frac{\partial^3 z}{\partial x^3} - 3 \frac{\partial^3 z}{\partial x^2 \partial y} + 4 \frac{\partial^3 z}{\partial y^3} = 0$ **03**
- (b) Solve the partial differential equation $\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$ by using the method of separation of variables. **04**
- (c) Find the solution of the one dimensional wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ satisfying the conditions $u(0, t) = u(L, t) = 0$; for $t > 0$, $u(x, 0) = \frac{\pi x}{L}$; for $0 < x < L$ and $u_t(x, 0) = 0$; for $0 < x < L$. **07**

Enrolment No./Seat No_____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-IV (NEW) EXAMINATION – WINTER 2024

Subject Code:3140610

Date:26-11-2024

Subject Name:Complex Variables and Partial Differential Equations

Time:02:30 PM TO 05:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	Marks
Q.1 (a) Find the principal argument of $z = \frac{-2}{1+i\sqrt{3}}$.	3
(b) Find the constants a, b, c if $f(z) = x + ay + i(bx + cy)$ is analytic.	4
(c) Show that $u(x, y) = x^2 - y^2 + x$ is harmonic. Find the corresponding analytic function $f(z) = u + iv$	7
Q.2 (a) Sketch the set $0 \leq \arg z \leq \frac{\pi}{4}$ and determine its domain?	3
(b) Find the continued product of all the values of $\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)^{\frac{3}{4}}$	4
(c) Find and sketch the image of the region $ z > 1$ under the transformation $w = 4z$.	7
OR	
(c) Find the bilinear transformation that maps the points $z_1 = 1, z_2 = i, z_3 = -1$ onto $w_1 = -1, w_2 = 0, w_3 = 1$ respectively. Find the image of $ z < 1$ under this transformation.	7
Q.3 (a) Evaluate $\int_C e^{\sin z^2} dz$, where C is $ z = 1$.	3
(b) Expand $f(z) = e^z$ in a Taylor series about $z = 0$.	4
(c) Evaluate $\int_C \frac{z}{(z-1)(z-2)^2} dz$, where C is $ z-2 = \frac{1}{2}$.	7
OR	
Q.3 (a) Classify the poles of $f(z) = \frac{1}{z^2 - z^6}$.	3
(b) Find the radius and region of convergence for $\sum_{n=1}^{\infty} \frac{z^n}{2^{n+1}}$.	4
(c) Expand $f(z) = \frac{1}{(z+1)(z-2)}$ in Laurent's series in the following regions: (i) $ z < 1$ (ii) $1 < z < 2$ (iii) $ z > 2$.	7

- Q.4** (a) Form a partial differential equation for the equation $z = ax + by + ct$ 3
- (b) Solve $xp + yq = 3z$. 4
- (c) Solve $(x^2 - yz)p + (y^2 - zx)q = z^2 - xy$. 7
- OR**
- Q.4** (a) Solve $z = px + qy - 2\sqrt{pq}$. 3
- (b) Form the partial differential equation of $z = f\left(\frac{x}{y}\right)$. 4
- (c) Solve $px + qy = pq$ by Charpit's method. 7
- Q.5** (a) Solve $\frac{\partial^2 u}{\partial x \partial y} = e^{-t} \cos x$. 3
- (b) Solve $(D^2 + 10DD' + 25D'^2)z = e^{3x+2y}$. 4
- (c) Using the method of separation of variables, solve $\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u$,
 $u(x, 0) = 6e^{-3x}$. 7
- OR**
- Q.5** (a) Solve $\frac{\partial^3 z}{\partial x^3} - 3\frac{\partial^3 z}{\partial^2 x \partial y} + 2\frac{\partial^3 z}{\partial y^3} = 0$. 3
- (b) Solve $\frac{\partial^3 z}{\partial x^3} - 2\frac{\partial^3 z}{\partial x^2 \partial y} = 2e^{2x}$. 4
- (c) Using separable variable technique, find the acceptable general solution to the one-dimensional heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ and find the solution satisfying the condition $u(0, t) = u(\pi, t) = 0$ for $t > 0$ and $u(x, 0) = \pi - x, 0 < x < \pi$. 7

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-IV EXAMINATION – SUMMER 2025****Subject Code:3140610****Date:17-05-2025****Subject Name:Complex Variables and Partial Differential Equations****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		MARKS
Q.1	(a) State: (i) Liouville Theorem and (ii) Cauchy Goursat Theorem	03
	(b) Solve $x(y - z)p + y(z - x)q = z(x - y)$	04
	(c) Verify that $u(x, y) = y^3 - 3x^2y$ is harmonic in the whole complex plane and find its harmonic conjugate $v(x, y)$	07
Q.2	(a) Solve $\frac{\partial^2 u}{\partial x \partial t} = e^{-t} \cos x$	03
	(b) Using residue theorem, evaluate $\int_c \frac{\cos \pi z^2}{(z-2)(z-1)} dz$ counterclockwise around the circle $c: z = 3$	04
	(c) Evaluate $\int_c \frac{z-2}{z(z-1)} dz$ where c is the circle $ z = 3$	07
	OR	
	(c) Evaluate $\int_c \frac{e^{2z}}{(z+1)^4} dz$ where c is the circle $ z = 2$	07
Q.3	(a) Solve $(p - q)(z - px - qy) = 1$	03
	(b) Discuss the convergence of $\sum_{n=1}^{\infty} \left(\frac{6n+1}{2n+5}\right)^2 (z - 2i)^n$ and also find the radius and circle of convergence	04
	(c) Find the Mobius transformation that maps $z_1 = 0, z_2 = -i, z_3 = 2i$ onto $w_1 = 5i, w_2 = \infty, w_3 = -\frac{i}{3}$ respectively. What are the invariant points of this transformation.	07
	OR	
Q.3	(a) Evaluate $\oint_c e^{z^2} dz$ where c is any closed contour. Justify your answer.	03
	(b) Find the image of the infinite strip $0 \leq x \leq 1$ under the transformation $w = iz + 1$. Sketch the region in the w - plane.	04
	(c) Prove that the n^{th} roots of unity are in geometric progression. Also, show that their sum is zero.	07
Q.4	(a) Identify the type of singularity of $f(z) = \frac{2-e^z}{z^3}$	03
	(b) Show that $\text{Log}(i^3) \neq 3 \text{Log } i$, for the principal branch	04
	(c) Solve $zpq = p + q$ by using charpit's method	07
	OR	
Q.4	(a) Find the residue of function $f(z) = \frac{2z+3}{(z+2)^2}$	03
	(b) Using CR equations, prove that $f(z) = \bar{z}$ is nowhere analytic	04

- (c) Determine the Laurent series expansion of $f(z) = \frac{1}{(z+1)(z+3)}$ valid for (i) $1 < |z| < 3$ (ii) $|z| > 3$ (iii) $0 < |z+1| < 2$ **07**
- Q.5** (a) Classify second order homogeneous partial differential equations as elliptic, parabolic or hyperbolic **03**
- i) $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial x \partial y} - 12 \frac{\partial^2 z}{\partial y^2} = 0$ ii) $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial x \partial y} = 0$
- (b) Solve $\frac{\partial^3 z}{\partial x^3} - 3 \frac{\partial^3 z}{\partial x^2 \partial y} + 4 \frac{\partial^3 z}{\partial y^3} = e^{x+2y}$ **04**
- (c) Using the method of separation of variable, solve $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$, $u(x, 0) = 6e^{-3x}$ **07**
- OR**
- Q.5** (a) Form a partial differential equation by eliminating the arbitrary function from $z - xy + f(x^2 + y^2)$ **03**
- (b) Solve $25r - 40s + 16t = 0$ **04**
- (c) Find the solution of the wave equation $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$ such that $y = a \cos pt$ when $x = l$, and $y = 0$ when $x = 0$ **07**